

ASIC Implementation of ECG Denoising FIR Filter by using Hybrid Vedic-Wallace Tree Multiplier

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Abstract. The design of hand-held portable devices for cardiovascular health monitoring based on the analysis of Electrocardiogram (ECG) is a hot topic of research nowadays. Digital filters, possibly designed using digital multipliers, are a critical module in the design of such analysis systems. This paper presents the ASIC design of a 64^{th} order digital low-pass FIR filter to be used in ECG denoising applications. The FIR filter is optimized for usage in battery-operated portable systems, wherein power consumption and on-chip area are crucial parameters of concern. These challenges are met by employing a novel multiplier architecture, which we design by hybridizing the two popular digital multiplication algorithms, namely the Vedic multiplication algorithm and the Wallace-tree multiplication algorithm. The proposed multiplier consumes lower area and power consumption than either of the aforementioned conventional multiplication algorithms. This is established through FPGA-based implementations on Artix-7 FPGA, for the various wordlengths. The area and power optimizations are further exercised by two proposed techniques, *viz.*, elimination of redundant multipliers, and power gating using the proposed zero-detector circuit. RTL-to-GDSII flow has been completed using Cadence digital design and sign-off tools for SCL-180 nm technology. The results indicate that the proposed filter architecture occupies 12.06% lower area and 75.63% lower power consumption than the contemporary designs described in the literature.

Keywords: ASIC Design, Biomedical Signal Processing, ECG Denoising, Hybrid Multiplier Design, Low Power FIR Filter, Vedic Multiplier, Wallace Tree Multiplier

1 Introduction

As per the World Health Organization's report, Cardiovascular diseases (CVDs) are responsible for around 17.7 million deaths across the world every year [1]. Despite all the advances in medical technology, Electrocardiography is still the first performed, universally accepted and widely used medical procedure for the diagnosis of CVDs [2, 3]. ECG is preferred because of its non-invasive nature and accuracy of diagnosis [4]. Early detection of CVDs is extremely crucial for a proper therapeutic cure. Therefore, in recent times, interest has grown for out-of-the-hospital continuous ECG monitoring devices [5]. Such devices are often battery-operated and need to be compact, imposing tight constraints on the area and power consumption of all the sub-system blocks [6].

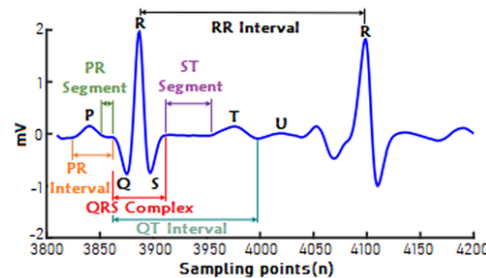


Fig. 1: ECG Signal Segments during a Typical Cardiac Cycle [7]

An ECG wave during a typical cardiac cycle along with the features is shown in Fig. 1. As observed from Fig. 1, the ECG signals contain a lot of pathological and physiological information which can be interpreted only by the study of its morphological and interval features. The morphological features include P-wave, QRS-complex, T-wave, and U-wave, while the interval features include PR interval, ST interval, RR interval, PR-segment, ST-segment, and so on [7].

During the acquisition of ECG signals from electrodes, the signals are heavily corrupted due to various noises. The presence of noise in ECG signals leads to difficulties in feature extraction, inaccurate estimation of characteristic points, etc., ultimately rendering them to be useless for clinical diagnosis [8]. ECG denoising (includes low pass filtering) is a critical step ahead of feature extraction and ECG analysis, wherein the noise sources are attenuated and ECG signals are accentuated [9]. Even though several software algorithms for ECG denoising exist, linear filter implementation on hardware is best suited for high-frequency artifact removal.

Area-constrained low-power filter architectures targeting ECG signals are scantily reported in the literature. In this paper, we propose an architecture of a low-pass FIR filter based on the Multiply-Accumulate (MAC) technique, exclusive for low-power ECG signal denoising. The proposed FIR filter is realized using a novel hybrid multiplier. It is well established that the various digital multiplier architectures differ in the schemes in which the partial products are reduced. Among all popular multiplication schemes, the literature suggests that the Wallace Tree scheme results in an area-efficient multiplier, while the Vedic Multiplication scheme results in power and energy-efficient multiplier [10]. In order to address both the power consumption and area constraints imposed by portable battery-operated systems, we propose to hybridize the two popular schemes, resulting in the aforementioned hybrid multiplier.

Our major contributions in this paper are as follows:

1. A novel architecture of the Hybrid Vedic-Wallace Tree Multiplier (HVWTM) is proposed.
 - The proposed multiplier HVWTM is compared against Vedic Multiplier (VM) and Wallace Tree Multiplier (WTM) on the basis of Artix-7 FPGA implementation results. It is established that the multiplier is both area and power optimised than the existing schemes of implementation.
 - The robustness of the proposed multiplication strategy is established through FPGA implementations for various wordlengths ($N= 4\text{-bit}, 8\text{-bit}, 16\text{-bit}$ and 32-bit).
2. Power gating of the proposed HVWTM is explored with the novel zero-detector circuit.
3. The number of multipliers required for FIR filter implementation is optimized by eliminating the redundant multipliers.
4. ASIC design based on RTL to GDSII flow is completed. Post-layout simulation is performed and the proposed FIR filter is compared against the contemporary research works.

The organization of this paper is as follows: Section 2 highlights the methodology used in carrying out the filter implementation and its verification. A brief review of contemporary works on FIR filter architectures in biomedical signal processing applications is discussed in Section 3. Section 4 describes the proposed multiplier architecture by hybridization of the two multiplication algorithms. Section 5 proposes the structure for 64^{th} order low pass FIR filter for ECG Denoising, followed by implementation results and discussion in Section 6.

2 Filter Implementation and Verification Methodology

This section is concerned with the strategic procedure to finalize filter design specifications and discussion on tools and techniques employed to implement and verify the architecture.

2.1 Target Specifications

In [11], the authors describe the detailed procedure to finalize the digital LPF specifications in low-area low-power constrained scenarios. This work follows the same approach. Accordingly, coefficients are

designed for 64^{th} order Blackman Windowing technique-based FIR filter. Numerous MATLAB simulations ascertain that the particular choice of filter design technique and filter order ensures the optimum balance between the Least Mean Square Error (MSE) between original and filtered signals, Improvement in Signal to Noise Ratio (SNR), and the hardware resources required for implementation. The frequency specifications for the FIR filter can be chosen only after studying the common ECG artifacts (represented in Table 1).

Table 1: List of Common ECG Artifacts

Sr. No.	Artifact	Origin	Frequency Range	Cause	LPF Suitable (Y/N)
1	Baseline Wander [12]	Physiological	0.05 to 1 Hz	Drift noise due to respiration activity of patient, body movements	N
2	fECG and mECG Artifacts [13,14]	Physiological	0.05 to 150 Hz	Presence of Fetal ECG in Pregnant Mother ECG and vice-versa	N
3	Muscular Artifacts (Electromyographic Noise) [12,15]	Physiological	20 to 1000 Hz	Muscular electrical activity due to shivering etc.	Y
4	Motion Artifacts [7]	Non-physiological	upto 1Hz	Poor electrode contact, Skin-electrode impedance	N
5	Power Frequency Interference [12,15]	Non-physiological	50-60 Hz	Capacitive and inductive coupling with power lines	N
6	Noise due to Nerve/Brain Stimulators, Pacemakers [16,17]	Non-physiological	100-160 Hz	Internal electrical artifacts due to presence of medical implant	Y
7	Electromagnetic Interference [18]	Non-physiological	800kHz to 1MHz	Electrical artifacts generated due to high magnetic field MRI machines in vicinity	Y
8	Instrumentation Noise [12,19]	Non-physiological	Around 100 Hz	Noise produced by cables, probes, Analog to Digital Converters	Y
9	Radio Frequency Interference [18]	Non-physiological	800kHz to 1MHz	Interference due to cell phones and walkie-talkies	Y
10	Channel Noise (e.g. AWGN) [7]	Non-physiological	*	Transmission of ECG through poor channel conditions	Y

The various ECG records in the MIT-BIH database [20] were considered and their frequency spectra are plotted in MATLAB. Let F_{max} be the frequency component beyond which the ECG signal amplitude is less than 1 mV, (i.e. the frequency components beyond F_{max} don't contribute significantly to the energy of the signal). The F_{max} for various ECG signal records is represented with a scatter plot in Fig. 2. Fig. 2 suggests that a cut-off frequency of 50 Hz is sufficient for filtering the noises reported in Table 1.

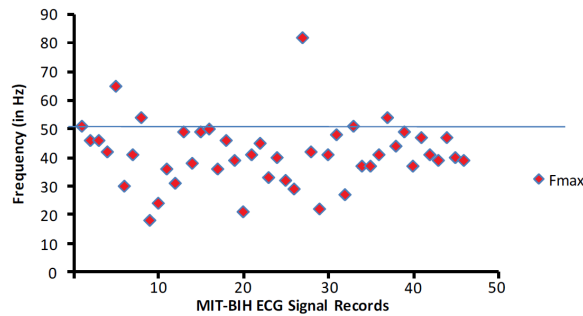


Fig. 2: Scatter Plot Representation of F_{max} for ECG Signal Records from MIT-BIH Database [20]

Most commercial ECG machines report use of 40 Hz as the cut-off frequency [21]. The AHA recommendations also consent to this cut-off frequency for non-ambulatory ECG monitoring. Therefore, the cut-off frequency is chosen as 50 Hz. Following the Nyquist criteria and the existing standards, a sampling frequency of 500 Hz is chosen. The resolution of 16-bits in modern ECG machines [22] is chosen in our proposed system design.

Based on aforementioned design considerations, the filter coefficients were computed using the Filter Design and Analysis (FDA) tool of MATLAB. The filter coefficients in standard exponential number format for various filter orders ($L = 16, 32$ and 64) are listed in Table 2.

Table 2: Filter Coefficients of the Proposed ECG De-noising FIR Filter

Filter Order, L	Filter Coefficients
16	$h(n) = \{5.685E-19, -6.849E-04, -2.243E-03, 1.452E-18, 1.722E-02, 6.06E-02, 1.268E-01, 1.901E-01, 2.165E-01, 1.901E-01, 1.268E-01, 6.06E-02, 1.722E-02, 1.452E-18, -2.243E-03, -6.849E-04, 5.685E-19\}$
32	$h(n) = \{1.623E-19, 2.743E-20, 1.955E-04, 8.124E-04, 1.677E-03, 1.899E-03, -1.342E-18, -5.168E-03, -1.287E-02, -1.919E-02, -1.73E-02, 5.203E-18, 3.619E-02, 8.744E-02, 1.421E-01, 1.842E-01, 2.004E-01, 1.842E-01, 1.421E-01, 8.744E-02, 3.619E-02, 5.203E-18, -1.73E-02, -1.919E-02, -1.287E-02, -5.168E-03, -1.342E-18, 1.899E-03, 1.677E-03, 8.124E-04, 1.955E-04, 2.743E-20, 1.623E-19\}$
64	$h(n) = \{-1.313E-19, 5.253E-06, -2.742E-20, -5.191E-05, -1.582E-04, -2.633E-04, -2.51E-04, 3.828E-19, 5.179E-04, 1.147E-03, 1.536E-03, 1.246E-03, -1.341E-18, -2.01E-03, -4.179E-03, -5.209E-03, -3.976E-03, 3.044E-18, 5.928E-03, 1.161E-02, 1.399E-02, 1.04E-02, -5.201E-18, -1.550E-02, -2.927E-02, -3.555E-02, -2.701E-02, 7.058E-18, 4.389E-02, 9.737E-02, 1.489E-01, 1.863E-01, 1.999E-01, 1.863E-01, 1.489E-01, 9.737E-02, 4.389E-02, 7.058E-18, -2.701E-02, -3.555E-02, -2.927E-02, -1.550E-02, -5.201E-18, 1.04E-02, 1.399E-02, 1.161E-02, 5.928E-03, 3.044E-18, -3.976E-03, -5.209E-03, -4.179E-03, -2.01E-03, -1.341E-18, 1.246E-03, 1.536E-03, 1.147E-03, 5.179E-04, 3.828E-19, -2.51E-04, -2.633E-04, -1.582E-04, -5.191E-05, -2.742E-20, 5.253E-06, -1.313E-19\}$

2.2 Tools and Techniques

The proposed architecture is implemented on Artix-7 FPGA *xc7a200tfbg676-2* platform for rapid prototyping by Xilinx. The particular FPGA based on the 28 nm high-K metal gate technology contains 215360 logic cells, 33650 logic slices, 740 DSP slices and operates at supply voltage 1.2V to 3.3 V [23]. Synthesis and implementation on the FPGA are carried out using Xilinx Vivado Design Suite 2019.2. Functional verification along with timing considerations is carried out at all the three stages, viz. RTL level, post-synthesis and post-implementation.

For ASIC implementation, the RTL-to-GDSII flow has been completed using SCL 180nm technology pdk files. The Cadence digital design tools used are Genus (for synthesis), Innovus (for physical design), Tempus (for performing Static Timing Analysis), and Incisive (for observing time simulation waveforms). The Calibre tool by Mentor Graphics is used for performing DRCs. Similar to FPGA, verifications are carried out at multiple stages.

The FDA tool of MATLAB is used in designing filter coefficients. This study also utilizes the ECG signal records from the MIT-BIH database available on Physionet[dot]org. The database contains 48 records of 30 minute long ECG signals from patients across gender and across various age groups recorded in laboratory conditions [24].

2.3 Functional Verification

To test the functionality of the proposed FIR filter, the sub-system modules were tested individually through test-bench verification. ECG records from the MIT-BIH database [24] were imported to MATLAB. Each of the raw ECG signal record was contaminated using high frequency synthetic noise ($f > 50$ Hz). Post this processing, the signal vectors with added noise are converted to test-vectors in text format, to be provided as input to the proposed HVWTM based FIR Filter. The file *io* options of VHDL allow the test-vectors to be read into the test-bench. The test-vectors are provided to the design and the output vectors are written in the output text file. The output vectors are then again imported to MATLAB and converted into signals for plotting. The filtered signal is then compared against the original signal taken from the database. The process was repeated for multiple ECG records, at post-synthesis and post-layout level.

3 Review of Recent Literature on Hardware Implementation of FIR Filter Architectures

This section presents a brief review of reported literature on the hardware implementation of FIR filters using various arithmetic computational logic circuits.

In early literature, FIR filter are implemented using pipelined carry propagate and carry-save multipliers and optimised using architecture folding [25], Common Subexpression Elimination (CSE) [26] and coefficient representation using Canonical Sign Digit (CSD) [27]. However, such implementations result in either low throughput per unit area or increased slice count [28,29]. In later works [30–32], Distributed Arithmetic (DA) based implementations are reported, but are not suited for ASIC implementation and result in higher slice count when implemented on FPGA [33].

Literature reports architectures optimizing the adder structure to enhance the overall performance. Techniques such as multiplexer based Carry Increment Adder (CIA) [34] and Branched-tree algorithm [35] are reported, targeting the combinational path delay. However, it must be pointed out that there is no critical requirement for high speed in filtering biomedical signals such as ECG. In [36–39], the architecture of transposed FIR filter is optimized using retiming transformation on delay elements. Delay-efficient adder and multiplier architectures have been proposed. Although the architectures result in higher throughput and energy efficiency, it imposes a penalty on the area.

An attempt toward lower power consumption is through the usage of reversible logic gates [40,41]. In [42], FIR filter architecture is proposed in which the adder (Square Root Carry Select Adder (sq-CSA) and the multiplier (VM) are realized using reversible logic gates. The power consumption is undoubtedly lowered but the drawback of the architecture is the large number of garbage outputs generated and associated fan-out limitations [43].

Contemporary research developments on FIR filter implementation propose Multiply-Accumulate (MA) based architectures. In Vedic multiplication algorithms, complexity of operation is lowered through parallel computation of partial products. The modular architecture results in much lower power consumption than other digital multiplier architectures [44,45]. In [10], authors have implemented 16th order FIR filters using various combinations of parallel-prefix adders and the multipliers (*viz.* VM and WTM). Across the choices of different filter coefficients (based on the filter specifications) and the choices of adder architecture, the results reported proving that the WTM-based architectures have improved performance and lower hardware resource utilization than VM-based architecture. However, the latter has lower power consumption and lower Energy Delay Product (EDP). Similar comparisons are reported in [46,47].

In [33], low pass filter architecture has been proposed for de-noising of ECG signals. The core of the proposed architecture includes Processing Element (PE) which is a Vedic multiplier incorporating Carry Look-Ahead Adder (VM-CLA) and a Control Unit(CU). The architecture uses ROM for the storage of filter coefficients and RAM for storage of input samples. The authors in [48] propose a similar PE-based filter architecture, however, they propose an improved strategy for partial product addition using Ripple Carry Adders(PPAVD-RCA). It can incontestably be argued that the limitation of these designs is the usage of off-chip RAM and ROM memories and a higher control path overhead. In [49], a wave-pipelined VM architecture is proposed to combat clock skew due to set-up and hold time violations. The signal arrival times without any pipelining are computed using CAD tools. Thereafter, based on the set-up and hold time violations reported, suitable delay elements between partial product generation and the final partial product addition stage are added to compensate for the delay of the VM. The resulting wave-pipelined VM-based FIR architecture can be operated at higher clock frequencies. However, this work doesn't report any discussion about power consumption, which is a more vital parameter in portable systems. In [50], a similar pipelining scheme is proposed for Wallace Reduction Tree based FIR filter architecture. The architecture results in very high throughput, but higher area occupancy.

4 Proposed VM-WTM Hybridization Strategy

Design of multiplier is most challenging task in the implementation of Digital Signal Processing (DSP) systems [51,52]. In [23], Power Performance Area (PPA) Cards of the various digital multiplier architectures have been compared. The results conclude that WTM occupies the least FPGA device resources (and area in case of ASIC implementation). On the other hand, VM consumes the least power and has the least energy density among all the digital multiplier architectures. The data path delay (and even the maximum frequency of operation F_{max}) of both VM and WTM are in close proximity. This section describes the proposed hybridisation strategy in partial product reduction, with the goal to design a hybrid multiplier architecture that should have low device utilization and at the same time consume less power.

4.1 Vedic Multiplication Algorithm

Numerous multiplication algorithms are proposed in Vedic mathematics, of which the Urdhva Tiryagbhyam Sutra is the universal algorithm for multiplying all kinds of numbers, without any condition on ranges of operands [53]. The multiplication algorithm decomposes each $N \times N$ multiplication into four $\frac{N}{2} \times \frac{N}{2}$ multiplications implemented using the same decomposition strategy. The strategy results in a highly modular structure, with the fundamental block being the 2×2 VM. The multiplications carried out in parallel result in higher speed and modular approach results in low power consumption [10,33,48].

4.2 Wallace Tree Multiplication Algorithm

The Wallace Tree algorithm strategizes to increase the speed of conventional long multipliers through optimal rearrangement of Half Adders (HAs) and Full Adders (FAs), in order to reduce the critical path. This scheme optimally reduces the number of partial product rows in multiple iterations. In the final iteration, the partial products are added using a conventional long adder [10,54]. This strategic arrangement not only enhances the speed of multiplication, but also the typical routing results in compact low area multiplier implementation [55].

4.3 Design of the HVWTM Architecture

If $A = a_{N-1}a_{N-2}...a_0$ is the N -bit multiplicand and $B = b_{N-1}b_{N-2}...b_0$ is the N -bit multiplier respectively, represented as:

$$A = \sum_{i=0}^{N-1} a_i 2^i \quad (1)$$

and

$$B = \sum_{j=0}^{N-1} b_j 2^j \quad (2)$$

Then the product P can be expressed as:

$$P = \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} a_i b_j 2^{i+j} \quad (3)$$

If $k = i + j$, then

$$P = \sum_{k=0}^{2N-1} p_k 2^k \quad (4)$$

where $p_k = a_i b_{k-i}$

In our proposed strategy, k^{th} parts of a and b are multiplied in a criss-cross wise manner using WTM and the resultant product terms p_k are added to arrive at the final product term P [56]. The steps involved are:

1. In the implementation of top module, The $N \times N$ multiplier and multiplicand are disintegrated into two $M \times M$ operands, where $M = \frac{N}{2}$, as per the VM strategy
2. All the four multiplications (viz. M MSBs of the multiplicand (A_U) with the M MSBs of the multiplier (B_V), the M LSBs of the multiplicand (A_V) with the M LSBs of the multiplier (B_V), 'M' MSBs of the multiplicand (A_U) with the M LSBs of the multiplier (B_V) and vice-versa) are performed with WTM. These generate the *vertical* and *crosswise* terms.
3. The four sets of partial products generated from the two *vertical* terms and the two *crosswise* terms are added using parallel adders, as in the case of VM.

It must be noted here that 4 X 4 HVWTM architecture is identical to that of 4 X 4 VM. This arrives from the fact that 4 X 4 HVWTM architecture would involve 2 X 2 WTM architectures. The 2 X 2 WTM and 2 X 2 VM are identical. Therefore, the hybridisation strategy for partial product reduction in case of 8 X 8 multiplication (for sake of illustration) is shown in Fig. 3.

Top Module: Vedic Multiplier

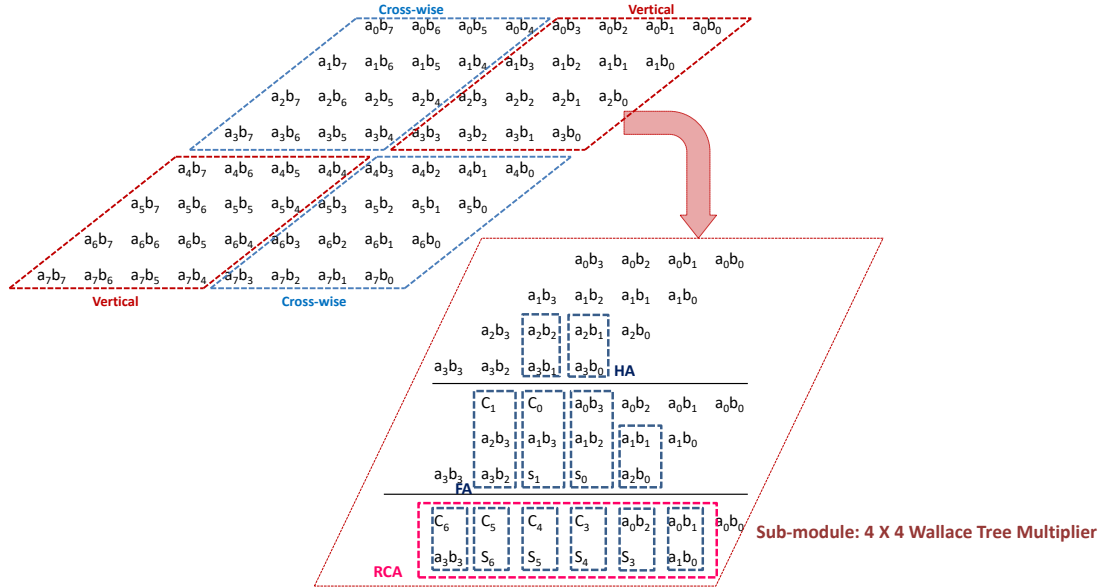


Fig. 3: Proposed Partial Product Reduction Strategy for 8 X 8 Multiplication

The conceptual representation of the proposed Hybrid multiplier architecture, HVWTM, is shown in Fig. 4, where M represents $\frac{N}{2}$. We implemented N-bit parallel adder based on ripple carry addition algorithm.

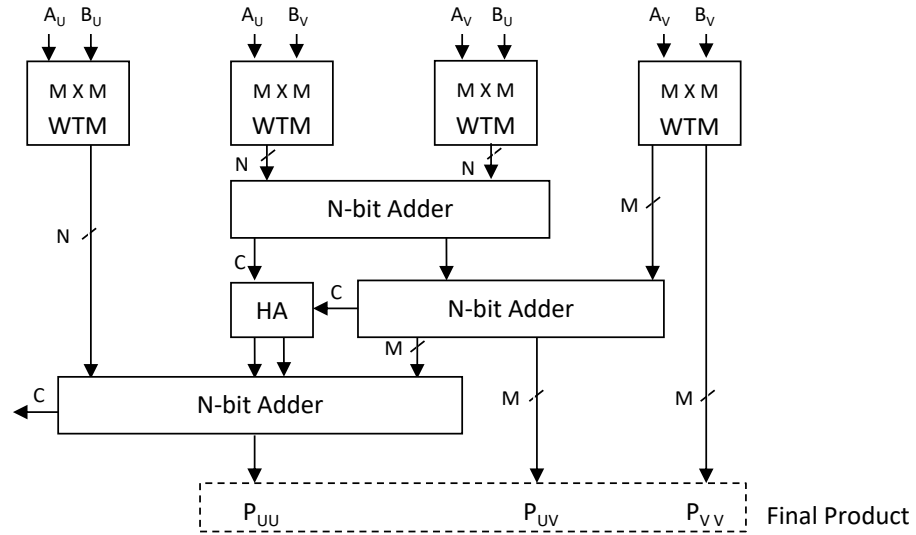


Fig. 4: Architecture of Proposed N X N HVWTM

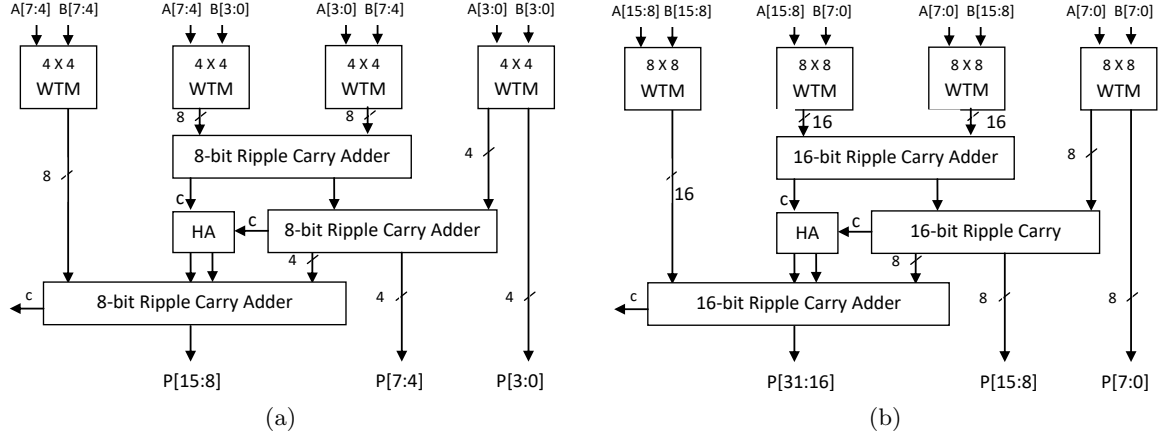


Fig. 5: Proposed Architecture of (a) 8 X 8 HVWTM and (b) 16 X 16 HVWTM Multiplier Architectures

Based on the concept described and represented in Fig. 4, the multiplier architectures of 8 X 8 HVWTM and 16 X 16 HVWTM are derived and are represented in Fig. 5a and 5b respectively. In the design of the proposed FIR filter, 16 X 16 HVWTM multiplier is employed.

Example Let $A = 11011011$ and $B = 10110010$ be the two 8-bit numbers to be multiplied. The scheme of multiplication of these numbers with the proposed HVWTM architecture is explained in Fig. ??.

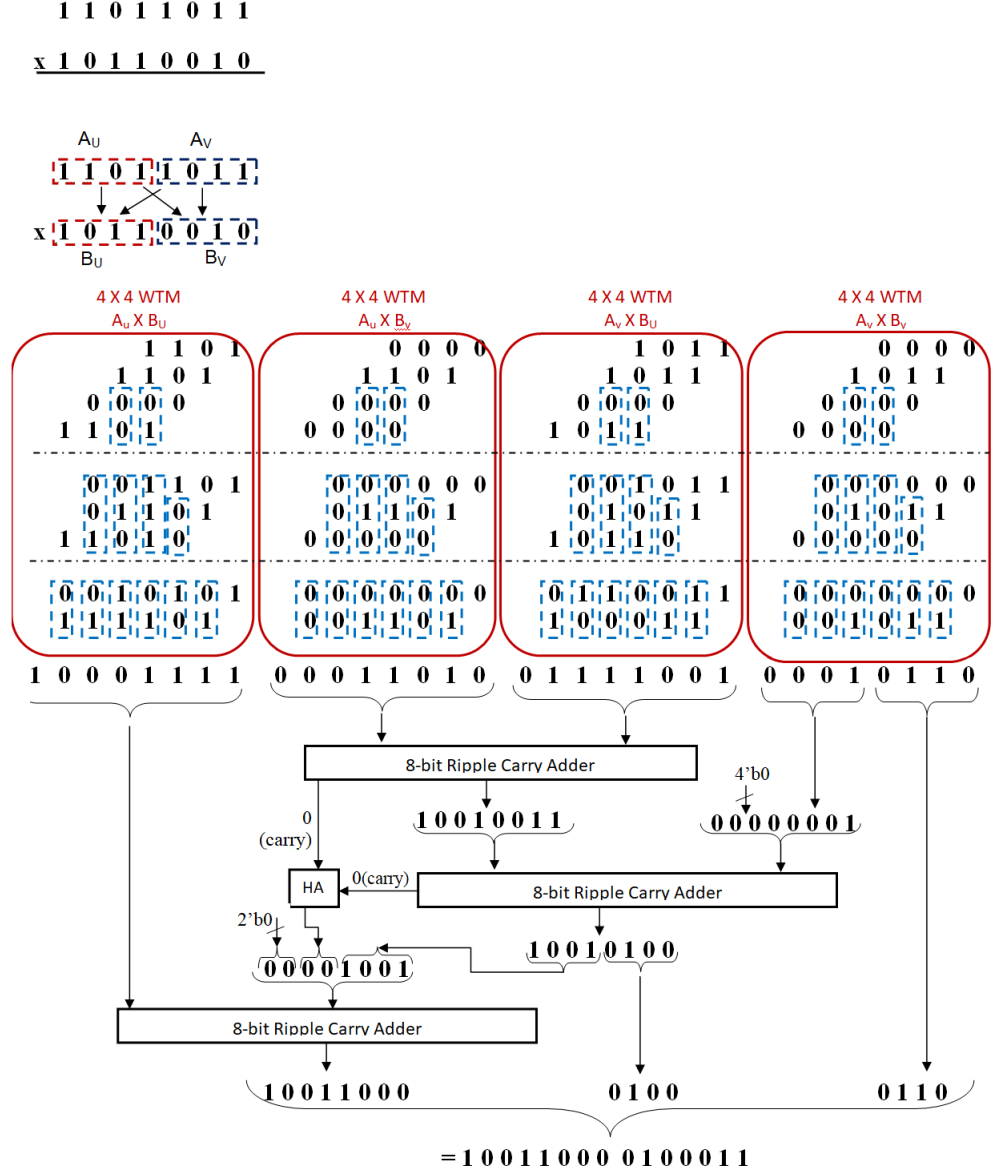


Fig. 6: Illustration of Scheme of Multiplication by Proposed HVWTM Architecture

In the proposed scheme, the both multiplicand A and multiplier B are disintegrated into two sets of 4-bits ($A_U = '1101'$, $A_V = '1011'$, $B_U = '1011'$, and $B_V = '0010'$). The top level multiplication is carried out by Urdhva Triyagbhyam Sutra (Vedic multiplication algorithm), while lower hierarchy multiplications are executed using the Wallace scheme. The sample simulation waveforms for checking the functionality of the proposed 16 X 16 HVWTM (in Xilinx Vivado 2019.2) is shown in Fig.7.

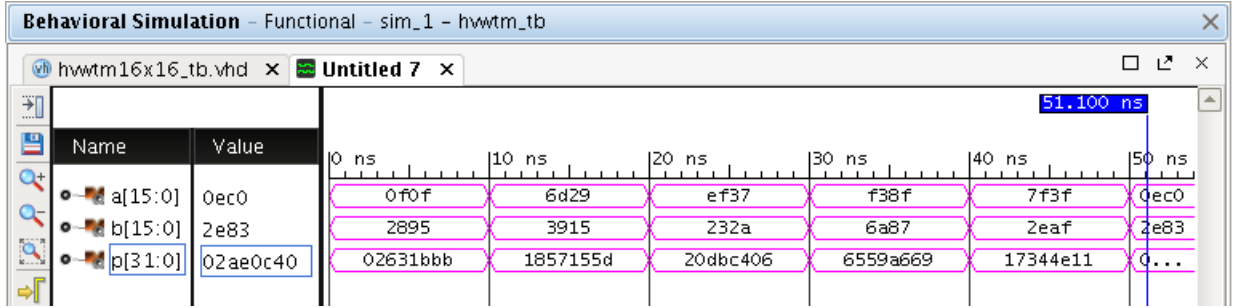


Fig. 7: Functional Verification of 16 X 16 HVWTM Multiplier Architecture

4.4 Theoretical Considerations in the Design of Proposed HVWTM Multiplier Architecture

The partial product reduction strategy proposed by Wallace [57] focussed on increasing the multiplication speed of the multiplier. The improvements to the Wallace scheme in successive WTM architectures have majorly targeted the critical path optimization. Hence, WTM is widely considered to be speed optimized multiplier design [58]. However, it is well understood from the architecture of WTM that as the wordlength increases, the number of partial products and the size of Wallace Tree also increase, resulting in longer interconnects [59]. While drawing the layout for ASIC implementation, longer interconnects essentially imply an increase in power consumption [60]. The capacitance associated with these interconnects connecting the functional blocks together adds up to the overall capacitance of the architecture, thereby increasing its dynamic power consumption [51]. It is estimated that 50% of dynamic power consumption is due to interconnects [61–63]. On the other hand, VM architecture because of its modularity enjoys the advantage of shorter interconnects [44, 45]. The architecture of VM is such that even when wordlength increases, the length of interconnects is not increased proportionately [23]. WTM is an area-optimized architecture with its implementation consuming the minimum number of HAs and FAs compared to other digital multiplier architectures [23]. VM architecture, despite being power-efficient, consumes more hardware resources. This can be attributed to more number of HAs required for implementation, which essentially do not reduce the number of partial product bits. Moreover, being a modular architecture designed in a bottom-up approach, VM ends up in a large number of HAs due to the compounding effect.

Table 3: Number of HAs and FAs required for Multiplier Implementation

Wordlength (N bits)	VM		WTM		HVWTM	
	#HA	#FA	#HA	#FA	#HA	#FA
4-bits	14	6	4	8	14	6
8-bits	65	39	16	47	25	41
16-bits	275	187	80	225	64	219

These are critical considerations in the design of the proposed HVWTM architecture. The proposed HVWTM architecture uses WTM for the implementation of lower wordlength blocks. This benefits in minimization of HAs and FAs, and ultimately the hardware resources required for implementation are

optimized. It can be deduced from Table 3 that as the wordlength increases, the number of HAs and FAs required for HVWTM implementation are in close proximity with those required for WTM implementation. At the same time, the top module (higher wordlength architecture) exploits the VM strategy of partial product reduction, resulting in the drawing of shorter interconnects compared to WTM. This ultimately results in much lower power consumption than WTM.

4.5 FPGA Implementation Results of the Proposed HVWTM architecture

Through experimentation, it is proved that both device utilization and power consumption could be lowered by the proposed hybridisation strategy. The performance parameters of HVWTM are compared against those of VM and WTM. The results of Device Utilization, Timing Summary and Power Consumption (for FPGA Implementation, described in Section 2.2) are reported in Table 4, 5 and 6 respectively.

From Table 4, it is evident that the HVWTM occupies lesser logic LUTs and slices than VM for all wordlength. The device utilization is close to that of WTM, for all wordlengths. It is seen that for $N = 16$ -bits, HVWTM occupies just 2.47% lesser LUTs and 1.8 % more slices than WTM. But it occupies 20.71% lesser LUTs and 17.91% lesser slices than VM. Similarly, for $N = 32$ -bits, the hybrid multiplier HVWTM occupies 25.08% lesser LUTs than VM. In essence, this indicates that the hybrid multiplier has inherited area-efficiency of WTM and is a more compact structure than VM.

Table 4: Device Utilization Comparison of VM, WTM and HVWTM

Architecture	Wordlength (N bits)	Number of LUTs (logic)	(% Utilization)	Number of Slices	(% Utilization)	Number of Bonded IOBs
VM	4-bits	17	0.01	5	0.01	16
	8-bits	108	0.08	30	0.09	32
	16-bits	449	0.34	134	0.4	64
	32-bits	1962	1.71	570	1.47	128
WTM	4-bits	12	<0.01	3	<0.01	16
	8-bits	98	0.07	28	0.07	32
	16-bits	365	0.27	108	0.32	64
	32-bits	1642	1.23	474	1.42	128
HVWTM	4-bits	17	0.01	5	0.01	16
	8-bits	85	0.06	23	0.07	32
	16-bits	356	0.27	110	0.33	64
	32-bits	1470	1.1	422	1.26	128

Table 5: Timing Summary Comparison of VM, WTM and HVWTM

Architecture	Wordlength (N bits)	Data Path Delay (in ns)	Logic Delay (in ns)	Route Delay (in ns)	WNS (in ns)	WHS (in ns)	Fmax (in MHz)
VM	4-bits	4.306	2.769	1.537	80.291	0.156	50.74
	8-bits	4.672	2.873	1.799	79.933	0.254	49.83
	16-bits	9.798	2.842	6.956	74.889	0.117	39.82
	32-bits	36.039	4.776	31.263	62.987	0.281	27.01
WTM	4-bits	4.509	2.863	1.646	80.087	0.141	50.22
	8-bits	4.567	2.767	1.799	80.039	0.232	50.10
	16-bits	10.112	2.755	7.357	74.510	0.141	39.23
	32-bits	30.466	4.348	26.118	68.206	0.190	31.45
HVWTM	4-bits	4.306	2.769	1.537	80.291	0.156	50.74
	8-bits	4.564	2.767	1.797	80.041	0.128	50.10
	16-bits	10.062	2.754	7.308	74.553	0.226	39.30
	32-bits	28.947	3.460	25.487	70.006	0.120	33.34

In Table 5, The maximum operable frequency for given multiplier under the slack constraints has been calculated as $F_{max} = \frac{1}{ActualClockPeriod - WNS}$ [23]. It is clear that the datapath delay and F_{max} values of HVWTM are in close proximity with those of VM and WTM. It is worthwhile to note that, as the wordlength increases, the Hybrid multiplier HVWTM can be operated at faster speeds than both VM

and WTM. For $N = 32$ bits, HVWTM can be operated at 33.34 MHz, which is marginally better than WTM ($F_{max} = 31.45$ MHz) and much better than VM ($F_{max} = 27.01$ MHz).

Table 6: Power Consumption Comparison of VM, WTM and HVWTM

Architecture	Wordlength (N bits)	Static Power Consumption (in mW)	Signal Power Consumption (in mW)	Logic Power Consumption (in mW)	IO Power Consumption (in mW)	Design Power Consumption (in mW)	Total Power Consumption (in mW)
VM	4-bits	0.139	0.105	0.073	2.548	2.726	2.865
	8-bits	0.161	0.653	0.604	6.674	7.931	8.092
	16-bits	0.317	4.323	3.129	15.686	23.138	23.455
	32-bits	1.674	13.521	11.041	33.934	58.496	60.169
WTM	4-bits	0.140	0.098	0.047	2.931	3.076	3.216
	8-bits	0.201	1.095	0.994	11.695	13.784	13.985
	16-bits	0.625	5.888	3.583	25.456	34.927	35.552
	32-bits	1.674	30.464	23.635	57.385	111.484	113.158
HVWTM	4-bits	0.139	0.105	0.073	2.548	2.726	2.865
	8-bits	0.169	0.689	0.552	8.119	9.360	9.529
	16-bits	0.292	3.558	2.540	15.486	21.584	21.876
	32-bits	1.674	12.205	9.116	35.997	57.318	58.992

Table 6 reveals that, for any given wordlength N , the power consumption of HVWTM is lower than that of WTM and close to that of VM. It is clear that power consumption of HVWTM is almost the same as VM. For $N = 8$ bits, the power consumption of HVWTM is 15.08% more than that of VM, while it is 6.73% less than that of VM for $N = 16$ bits. But when compared with WTM, the power consumption of HVWTM is reduced by 31.86%, 38.47%, and 47.87% for $N = 8$ bits, 16 bits and 32-bits respectively. Therefore, HVWTM has inherited the power efficiency of VM.

Robustness of the Results: Although the filter specifications require usage of $N = 16$ bits multiplier only, the results are reported for $N = 32$ bits as well to explore the robustness of the proposed hybridisation concept. The results are both wordlength and platform independent. Concluding, we can imply that by the said hybridization strategy, we are able to lower the device utilization of the multiplier and attain it close to WTM, while its power consumption is also lowered and is close to VM.

4.6 Optimization of the Proposed 16 X 16 HVWTM using Power Gating

Power gating is an important approach to reduce power consumption in a chip, wherein the blocks not performing operations are turned-off temporarily, so that the switching actions of the transistors in the region is minimized. Lower switching implies lower dynamic power consumption.

In a multiplier, if any of the operand is zero, then the resultant product is zero. In MAC based FIR filter, each multiplier has two inputs i.e. signal samples and filter coefficients. If either the signal samples or filter coefficients are zero and this knowledge is known beforehand, the product could be deduced as zero without having the transistors to switch values. To implement this logic, a zero-detector circuit is required. Such a circuit proposed by Saha *et al.* [56] consists of OR gates only. This circuit requires 90 transistors. The circuit is shown in Fig. 8(a). However, we propose NOR-AND-NAND gates based three layered Zero-detector circuit, in which the number of transistors required for implementation is reduced to 72. The proposed zero detector circuit shown in Fig. 8(b), which is employed to gate the HVWTM.

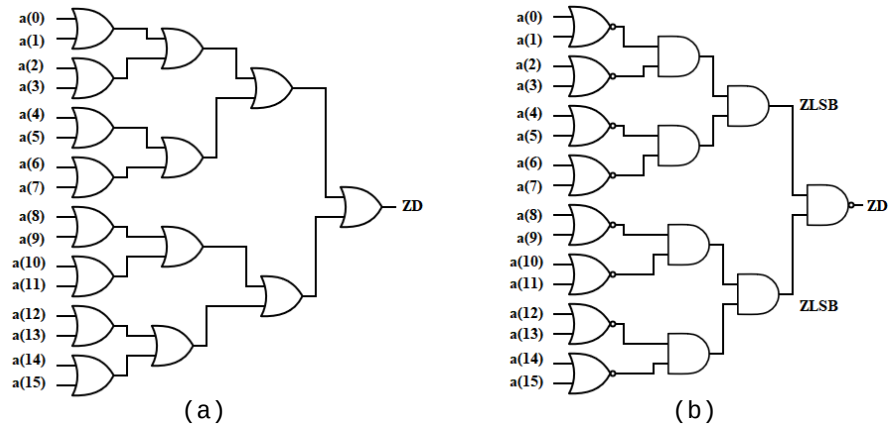


Fig. 8: 16-bit Zero Detector Circuit (a) Proposed in [56] (b) Our Proposed Design

Let A and B be the two 16-bit inputs to the multiplier. Let XX denote that both LSB (Least Significant Byte) and MSB (Most Significant Byte) of the 16-bit input are non-zero sequences, X0 indicates the LSB of the input is zero, while 0X indicates that MSB of the input is zero. The following signals are then defined as:

1. ZMSBA: MSB of is A zero (0X)
2. ZMSBB: MSB of is B zero (0X)
3. ZLSBA: LSB of is A zero (X0)
4. ZLSBB: LSB of is B zero (X0)

These signals can be generated with the same zero detector circuit shown in Fig. 8(b). The possible combinations and their corresponding Product P is represented in Table 7.

Table 7: Zero Detector Circuit based Optimisation Scheme

A	B	ZMSBA	ZMSBB	ZLSBA	ZLSBB	P
XX	XX	0	0	0	0	XXXX
XX	X0	0	0	0	1	XXX0
X0	XX	0	0	1	0	XXX0
X0	X0	0	0	1	1	0000
XX	0X	0	1	0	0	0XXX
XX	00	0	1	0	1	0000
X0	0X	0	1	1	0	0XXX
X0	00	0	1	1	1	0000
0X	XX	1	0	0	0	0XXX
0X	X0	1	0	0	1	0XXX
00	XX	1	0	1	0	0000
00	X0	1	0	1	1	0000
0X	0X	1	1	0	0	00XX
0X	00	1	1	0	1	0000
00	0X	1	1	1	0	0000
00	00	1	1	1	1	0000

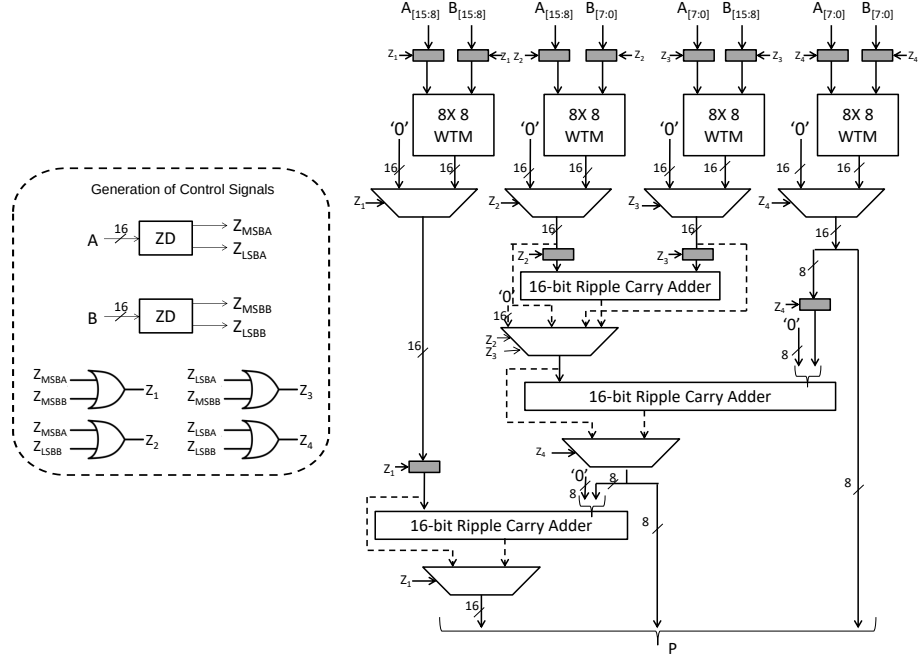


Fig. 9: Power Gating of Submodules of 16 X 16 HVWTM Architecture

Table 7 reveals that if there is a sequence of zeros at the MSB or LSB, then the product also contains a sequence of zeros. The top module of the proposed HVWTM is a modular structure, wherein each 16 X 16 HVWTM is built of four 8 X 8 WTM. Therefore, based on these control signals, each of the four 8 X 8 WTM are individually gated. The gated circuit is shown in Fig. 9. In Fig. 9, the shaded rectangles represent level-triggered latches. The sequences of 0's in MSBs or LSBs in the multiplier and multiplicand are computed to generate the control signals. The latches ensure that, as long as the sequences of 0's remain in the inputs, the changes in the inputs are not transparent to the particular multiplier or adder. This ultimately lowers the number of signal transitions in the computational blocks, which in turn reduces the power consumption of the entire architecture.

Table 8: FPGA Implementation Results Comparison of HVWTM and Optimised HVWTM (Power-gated)

Architecture	Wordlength (N bits)	Number of LUTs (logic)	Number of Slices	Data Path Delay (in ns)	F_{max} (in MHz)	Static Power Consumption (in mW)	Total Power Consumption (in mW)
HVWTM	16-bits	356	110	10.062	39.30	0.292	21.876
Optimised HVWTM	16-bits	483	150	5.612	47.62	0.132	4.211

Table 8 summarises the optimisation brought by the optimised HVWTM in terms of device **utilization**, timing summary and power consumption. Although, the control circuitry increases the device **utilization**, the power consumption is lowered by five folds. Quantitatively, for $N = 16$ bits, power gating increases the number of LUTs by 26.29% and number of slices by 36.36%, the overall power consumption is lowered by a massive 80.75%. This essentially proves the significance of power gating on the overall DSP architecture.

5 Design of Proposed Low Pass FIR Filter Architecture

Literature suggests that FIR filters are feed-forward systems that can be implemented as Multiply and Accumulate (MAC) units. In such implementation, the input samples are shifted through a tapped delay-line. Each shifted sample is multiplied by the corresponding filter coefficient and the products are accumulated. Therefore, this implementation demands as many multipliers and adders as the number of filter taps. If $X_{in}(n)$ denotes input samples stream to the FIR filter, then for a L^{th} order filter, the output response $Y_{out}(n)$ is defined as [64]:

$$Y_{out}(n) = \sum_{k=0}^{L-1} h(k)X_{in}(n-k) \quad (5)$$

where $h(k)$ is the impulse response for k^{th} instance (where $0 \leq k \leq L$)

A necessary requirement for linear-phase response is that the FIR filter must have symmetric filter coefficients. In other words, the impulse response $h(n)$ of the FIR filter must satisfy [10]:

$$h(n) = \pm h(L-n-1) \quad (6)$$

where L is the filter order and $n = 0, 1, 2, \dots, (L-1)$

From Equation 6, each time-shifted input signal sample in the tapped delay-line would be multiplied by the same-value filter coefficient twice, as the filter coefficients repeat. However, by exploiting the symmetric nature of filter coefficients, the number of multipliers required to implement the FIR filter are reduced to half compared to the conventional Direct Form-I implementation. The output response is defined as:

$$Y_{out}(n) = \sum_{k=0}^{\frac{L}{2}-1} h(k)(X_{in}(n-k) + X_{in}(L-1-k)) \quad (7)$$

The architecture of the symmetric linear-phase low-pass Vedic FIR filter designed using 16 X 16 HVWTM is shown in Fig. 10. In accordance with Equation 7, the redundant multipliers have been eliminated. Therefore, the architecture implementation results in lower area and power than the conventional Direct Form-I implementation. The quantitative analysis of the filter implementation follows in subsequent section 6.

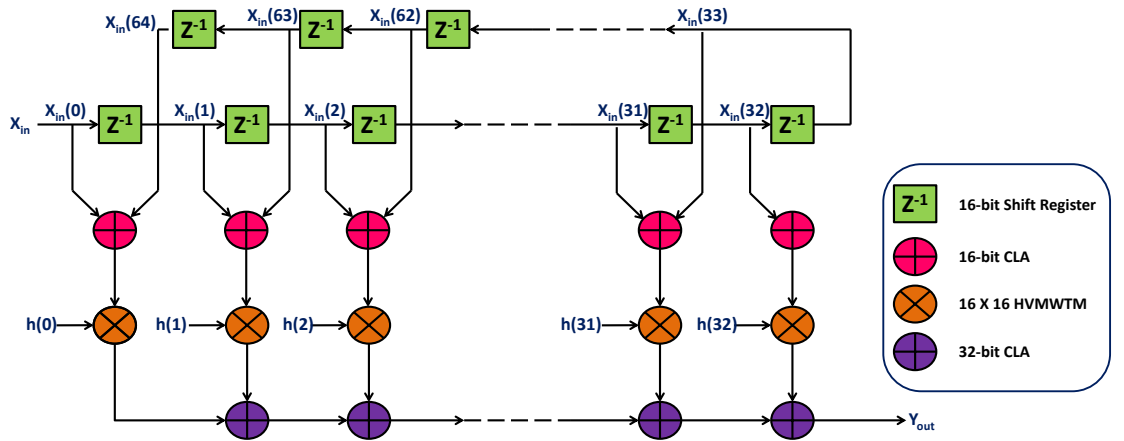


Fig. 10: Proposed Architecture of the 64th order FIR Filter

6 Results and Discussion

This section is aimed to highlight few of the salient advantages of the proposed HVWTM-based FIR filter architecture, by comparing it against the state-of-the-art FIR filter implementation (available in the literature). The study is done for implementation results for both FPGA and ASIC. The details are discussed in the subsequent subsections.

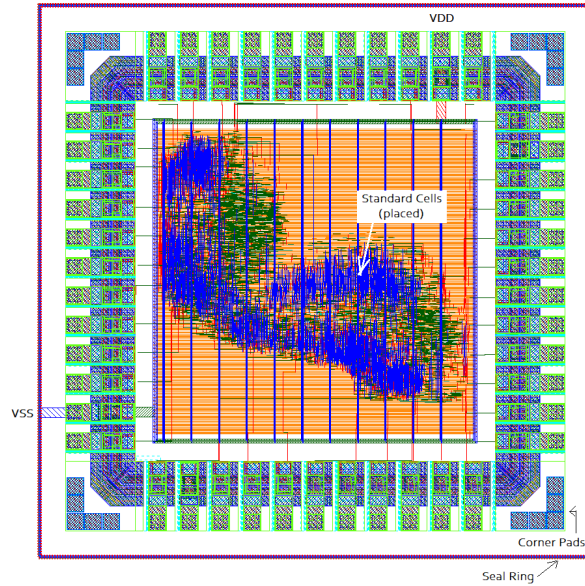


Fig. 11: Chip Layout of the Proposed HVWTM-based FIR Filter Architecture

The chip-layout of the proposed HVWTM-based FIR filter is shown in Fig. 11. The entire physical design is completed using the Innovus tool of Cadence Digital Design Suite. Seal-ring is placed and dummy are filled as per requirements given by Semiconductor Lab (SCL), Mohali. The chip has been sent for tape-out. The testbench verification (for post-layout simulation) of the proposed HVWTM-based FIR filter is illustrated in Fig. 12.

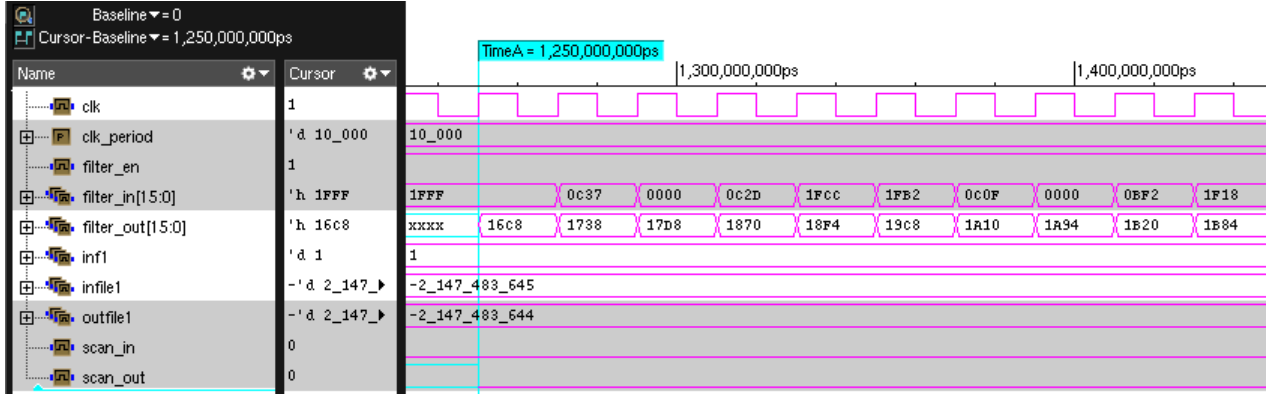


Fig. 12: Simulation Waveform for Testbench Verification of the Proposed HVWTM-based FIR Filter

6.1 Functional Verification

As described in subsection 2.3, simulation based functional verification is performed on the Record 101 of the MIT-BIH database [24]. Fig. 13a shows few QRS complexes of the ECG signal record. The signal is contaminated with high frequency synthetic noise shown in Fig. 13b. The resultant noisy signal is shown in Fig. 13c.

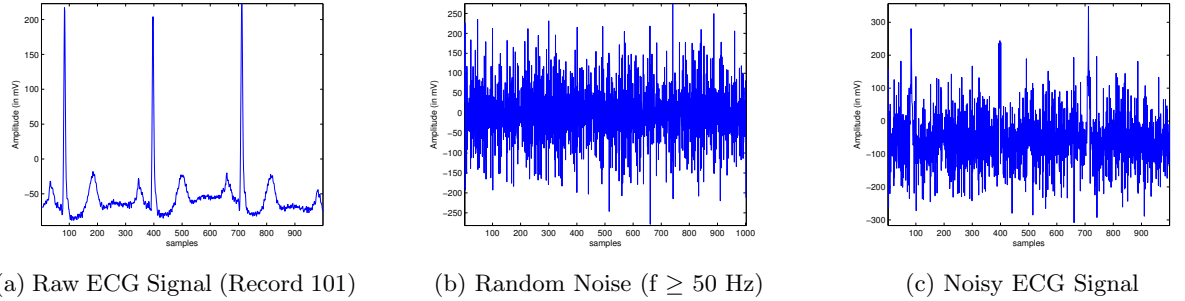


Fig. 13: Illustration of Addition of Random Noise

The noise contaminated ECG signal is filtered using the Proposed HVWTM-based FIR filter (Fig. 10). The ECG denoising is represented in Fig. 14, where the filtered ECG signal has been plotted against the noisy ECG. Features are recovered from the heavily noise-contaminated signal.

The filtered ECG signal is then compared against the Raw ECG signal from the database in Fig. 15. The proposed HVWTM-based FIR filter rejects noise and retains all the crucial features of the ECG signal. However, a linear phase shift is introduced due to the feed-forward nature of the FIR filter architecture. Also, the filtered signal is attenuated compared to the raw ECG signal. The output bits are shifted left by 1 bit-position to minimise the effect of the attenuation introduced in the system. **It is worthwhile to note that feature extraction from ECG signals usually involves analysis in time domain and amplitude of the signal holds only a ancillary importance in identification of various ECG waves (P-wave, QRS Complex, S-wave and T-wave). Nevertheless, we studied the impact of signal attenuation with different ECG samples from MIT-BIH database [24]. It was observed that all the waves of the ECG signal were clearly identifiable.**

Fig. 16 illustrates ECG De-noising by various filter orders ($L=16, 32$ and 64) of the proposed HVWTM-based FIR filter. Higher-order filters have greater linear phase shift and a higher degree of attenuation. As a matter of observation, the output response samples are shifted by a number half the filter order (i.e. for filter order $L=16, 32$ and 64 ; the samples in output response are shifted by $8, 16$ and 32 samples respectively).

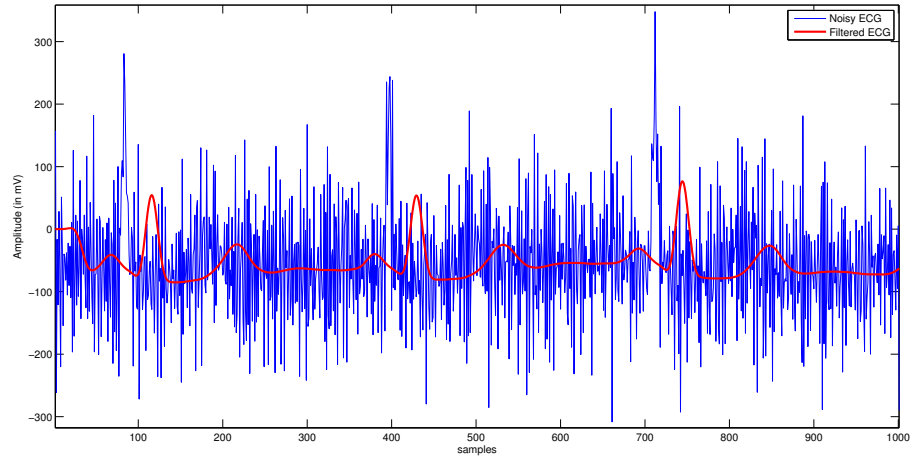


Fig. 14: Illustration of ECG-Denoising by Proposed HVWTM-based FIR Filter

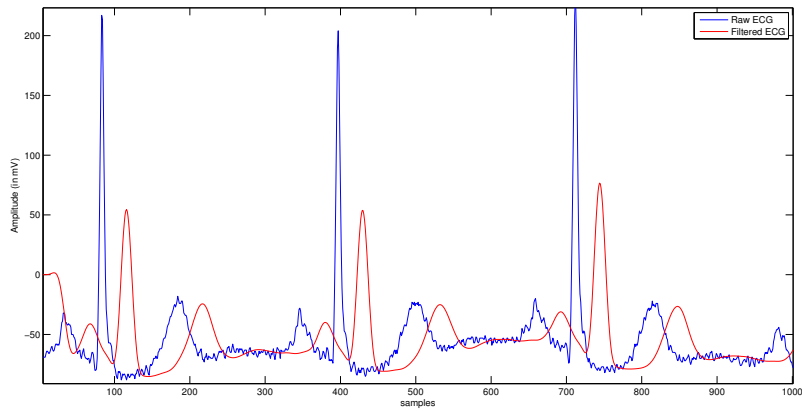


Fig. 15: Comparison of Raw ECG Signal with the Filtered ECG Signal (Record 101 of MIT-BIH Database)

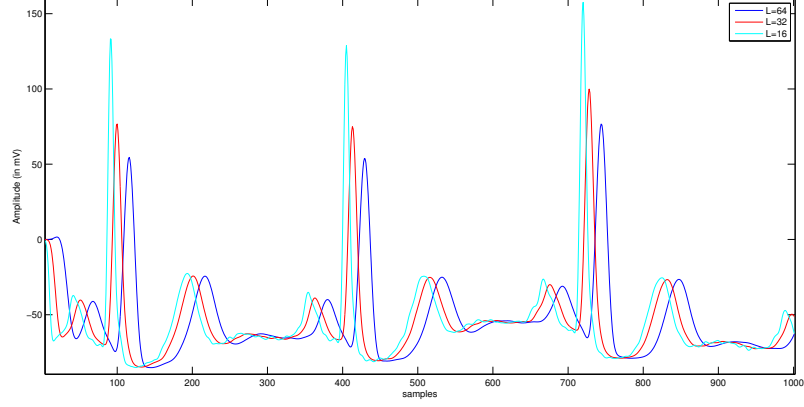


Fig.16: Effect of Filter Order of Proposed FIR Filter on ECG Denoising (Record 101 of MIT-BIH Database)

The experimentation is repeated for all the records of the MIT-BIH database [24] and for each record, Mean-Square Error (MSE) is computed between the uncontaminated raw ECG signal and the corresponding filtered ECG signal. The average of MSEs reflects 0.135 mV^2 , 0.133 mV^2 and 0.134 mV^2 for $L=16$, 32 and 64 respectively.

6.2 Energy Profile Parameters

It is also mandatory to compare the architectures based on energy parameters Energy of Operation (EOP) and Energy Throughput (ET) [23]. EOP is a crucial metric which compares the functionally equivalent architectures on basis of their energy efficiency, i.e. the energy consumed to complete the same operation. Mathematically, it is represented as [23]:

$$EOP = \int_0^T W(t).dt \quad (8)$$

where T refers to #clock cycles required to complete one operation and $W(t)$ refers to the instantaneous power consumption

If the power consumption is assumed to be constant (W_{av}), then EOP can be expressed as [23]:

$$EOP = \frac{W_{av} * L}{f} \quad (9)$$

where L refers to the filter order and f is the operating frequency.

ET is defined as the quantity of energy dissipated per output bit generated. This metric is crucial as it combines the total data processed per clock cycle and the energy dissipation for it, into the same metric. This metric is therefore useful to compare architectures that are functionally equivalent but perform operations at different mathematical scales. Mathematically, ET is represented as [23]:

$$ET = \frac{EOP}{Q} \quad (10)$$

where Q is the number of output bits per cycle.

6.3 FPGA Implementation Results

The proposed FIR filter is compared with state-of-the-art FIR filter implementations in Table 9.

Table 9: Comparison of FPGA Implementation Results (at 10 MHz clock frequency)

	Technique	Filter Order, L	#Slices	#LUTs	Delay (in ns)	Power Consumption (in mW)	EOP (in nJ/op)	ET (in nJ/bit)
Mittal, Nandi and Yadav (2017) [10]	*VM-BKA	16	368	1242	35.444	89.274	142.838	8.927
		32	782	2686	44.491	165.081	528.256	33.016
		64	1595	5506	62.789	315.891	2021.696	126.356
Prasad, Geetha and Srinivasan (2019) [49]	Wave-pipeline	16	314	956	26.7	66.480	106.368	6.648
		32	699	2161	36.267	118.103	377.930	23.6206
		64	1363	4399	49.777	213.798	1368.307	85.519
Padmavathy, Saravanan and Vimalkumar (2020) [48]	**PPAVD-RCA	16	288	974	31.884	77.305	123.688	7.731
		32	653	2182	38.207	130.959	419.069	26.192
		64	1285	4382	58.570	242.458	1551.731	96.983
HVWTM based Proposed FIR Filter	Multiplier Hybridization	16	200	563	27.599	27.918	44.669	2.792
		32	447	1304	31.965	40.309	128.989	8.062
		64	778	2483	40.345	59.025	377.761	23.612

*VM-BKA: Vedic Multiplier Brent-Kung Adder **PPAVD-RCA: Partial Product Addition in Vedic Design- Ripple Carry Adder

It is evident from Table 9 that the proposed HVMWTM-based FIR filter architecture is the most economical design in terms of FPGA device utilization. The proposed FIR filter occupies 39.46% lesser slices than the design proposed in [48], for filter Order $L = 64$. The architectures in [10], [49] and [48] utilise VM resulting in a slighter higher LUT and slice count. A 43.34% reduction in the number of logic LUTs is observed compared to the design proposed in [48]. The FIR filter architectures proposed in [10, 48] utilize high-speed adder architectures such as parallel prefix adders. These structures although offer higher speeds, however, results in the consumption of more hardware resources. It is therefore well-established from the results that the proposed hybridisation strategy results in lower FPGA device utilization. Also, there is a significant improvement (18.95%) in design timing which also can be attributed to the hybridisation strategy. However, ECG-denoising circuitry is not expected to work at very high frequencies, as ECG in itself is a very low-frequency signal.

Table 9 reports multifold improvement in power consumption compared to the state-of-the-art architectures for all the filter orders. This reduction in power consumption is not only due to multiplier hybridisation, but also due to the power gating strategy adopted. The reduction in dynamic power consumption can be attributed to the lower number of transitions on the signals because of gating, resulting in overall reduction in total power consumption. Results indicate a massive 72.39% reduction in the power consumption (for filter order $L = 64$) compared to the design reported in [49].

The energy parameters EOP and ET are also improved in the proposed FIR filter architecture. For filter order $L = 64$, the improvement in EOP and ET is 72.44% and 72.39% respectively compared to the architecture reported in [49]. The FPGA implementation results of 64th order proposed HVWTM-based FIR filter are represented graphically in Fig. 17.

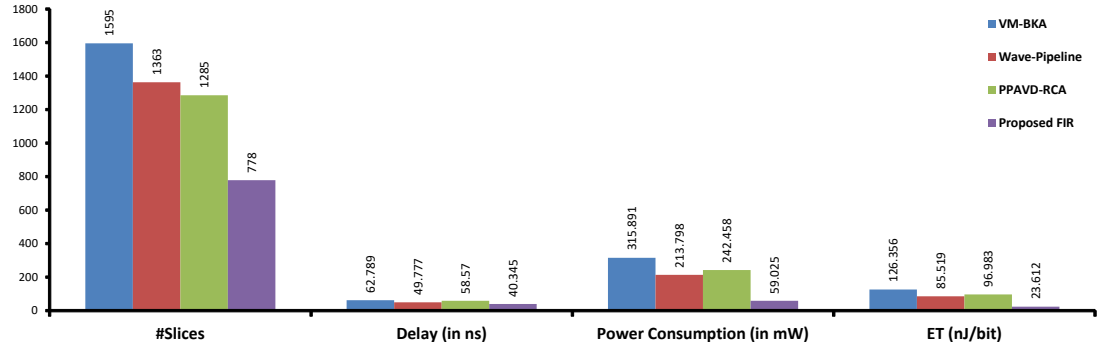


Fig. 17: Comparison of FPGA Implementation Results (at 10 MHz clock frequency)

In order to bring out the advantages of the proposed architecture through graphical viewing, Fig. 18 shows graphically the percentage improvement in critical parameters compared to the state-of-the-art literature.

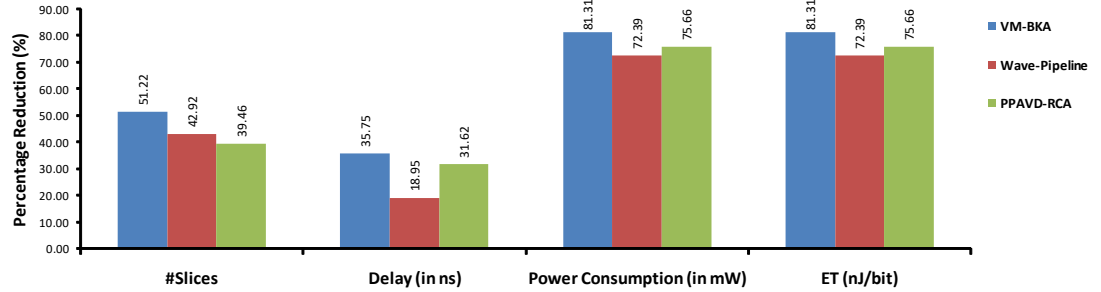


Fig. 18: Graphical Representation of Percentage Reduction in Critical Parameters for FPGA Implementation

6.4 ASIC Implementation Results

The experimentation is done with the ASIC design environment (mentioned in Section 2.2). The results are summarised in Table 10.

Table 10: Comparison of ASIC Implementation Results (at 100 kHz clock frequency)

Technique	Filter Order, L	Area (in μm^2)	Delay (in ns)	Power Consump-tion (in μW)	EOP (in pJ/op)	ET (in pJ/bit)
Mittal, Nandi and Yadav (2017) [10]	*VM-BKA	16	138805	41.965	375.338	600.542
		32	304530	47.885	722.469	2311.901
		64	600679	67.247	2291.911	14668.200
Prasad, Geetha and Srinivasan (2019) [49]	Wave-pipeline	16	145530	25.957	491.929	787.087
		32	315699	36.543	375.949	1203.041
		64	621130	50.713	2131.791	13643.500
Padmavathy, Saravanan and Vimalkumar (2020) [48]	**PPAVD-RCA	16	142418	45.472	364.890	583.825
		32	306457	46.197	936.517	2996.86
		64	590538	68.168	2041.799	13067.500
Patali and Kassim (2023) [66]	Approximate Computing	16	139938	20.402	292.785	468.455
		32	279876	28.723	585.569	1873.821
		64	559752	39.861	1171.138	7495.284
HVWTM based Proposed FIR Filter	Multiplier Hybridisation	16	114960	51.605	34.245	54.793
		32	250901	58.228	168.707	539.863
		64	492246	74.295	285.525	1827.37

*VM-BKA: Vedic Multiplier Brent-Kung Adder **PPAVD-RCA: Partial Product Addition in Vedic Design- Ripple Carry Adder

It can be deduced from Table 10 that there is 12.06% reduction in the area (for filter order, 64) compared to the design reported in [66], which is definitely the merit of the proposed design. Furthermore, there is an immense reduction in the power consumption of the design for all the wordlengths. For the 64th order FIR filter, there is a reduction of 75.63% in power consumption compared to the design reported in [66]. Also, there is 75.62% and 75.62% reduction in EOP and ET respectively, in comparison to the design reported in [66]. The ASIC implementation results for the 64th order FIR filter based on the proposed HVWTM is represented in Fig. 19.

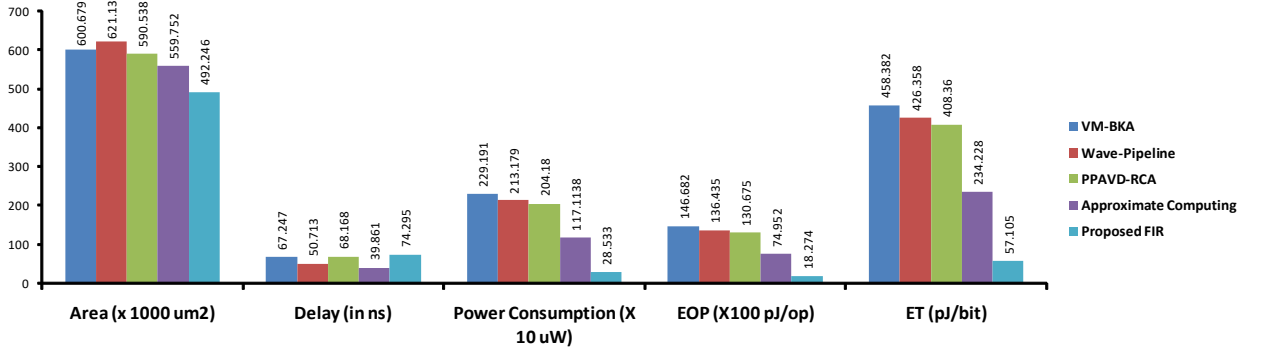


Fig. 19: Comparison of ASIC Implementation Results (at 100 kHz clock frequency)

The ASIC implementation results are in line with the FPGA implementation results and ascertain that the proposed multiplier hybridisation strategy results in lower on-chip area and power-consumption. The power consumption is further massively lowered due to the proposed power gating scheme.

Fig. 20 represents gains offered by the proposed solution over those described in the literature. Biomedical signals such as ECG don't have high speed requirements as they are very low frequency signals.

Therefore, the increase in delay is not much concerning, however, 75.63% reduction power consumption is a significant achievement.

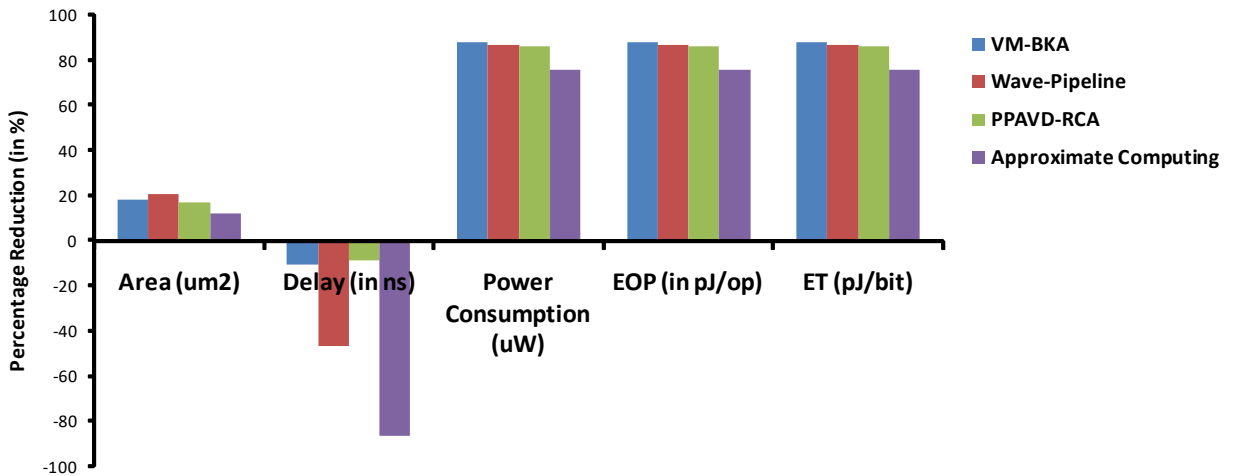


Fig. 20: Graphical Representation of Percentage Reduction in Critical Parameters for ASIC Implementation

7 Conclusion

ECG is a vital non-invasive cardiovascular diagnostic procedure. Denoising of ECG is a critical procedure, of which the low-pass filter is an important subsystem block. In DSP systems, the multiplier is the most critical block to be optimized. In this paper, we proposed a hybrid multiplier architecture, that is more efficient in both on-chip area and power than the conventional digital multiplier architectures. This is a critical requirement in portable biomedical signal processing systems. Also, power-gating of HVWTM is done, which immensely reduces power consumption. Validations on multiple records from MIT-BIH database and the ASIC implementation results indicate that the chip is suitable for applications in ECG beyond all doubts. Presently, the chip has been taped-out.

Acknowledgments

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